

Kwabena — perfect. Let's turn everything you did in Day 65 into a **clean, professional GitHub documentation page**. This is not a "summary paragraph." This is a **README-style project doc** — structured, technical, and portfolio-ready.

No screenshots except the topology (you can add that manually). All IPs typed out. Clear sections. Recruiter-friendly.

Here's your full Day 65 GitHub documentation draft:

Day 65 — Multi-Router Network Design, Routing, VLAN Correction & Troubleshooting

A three-router, three-LAN Packet Tracer lab demonstrating subnetting, dynamic routing (RIP v2), VLAN alignment, and multi-hop connectivity troubleshooting.

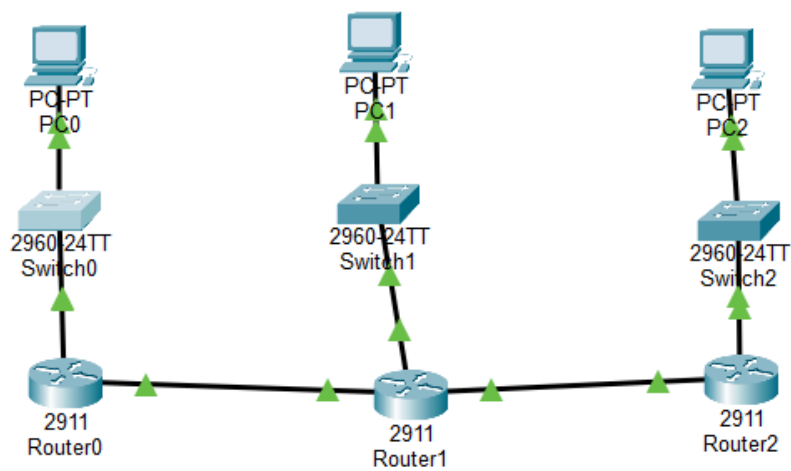
Project Overview

Day 65 introduces a realistic multi-router network topology similar to small enterprise environments. The goal is to design three separate LANs, connect them through three routers, enable dynamic routing, and verify end-to-end communication across all networks.

This lab combines:

- Subnetting
- Router interface configuration
- RIP v2 dynamic routing
- Switch VLAN configuration
- Multi-hop ping testing
- Troubleshooting broadcast domain isolation

Network Topology



Three routers connected in series:

Code

Router0 ↔ Router1 ↔ Router2

Each router connects to its own LAN:

Code

LAN 10 → PC0

LAN 20 → PC1

LAN 30 → PC2

1 2 3 4 Subnet Design

All LANs use /26 networks carved from a larger block.

LAN 10

- Network: 192.168.10.0/26
- Gateway (Router0 g0/0): 192.168.10.1
- PC0: 192.168.10.10

LAN 20

- Network: 192.168.20.0/26
- Gateway (Router1 g0/0): 192.168.20.1
- PC1: 192.168.20.10

LAN 30

- Network: 192.168.30.0/26
- Gateway (Router2 g0/0): 192.168.30.1
- PC2: 192.168.30.10

Subnetting ensures each LAN has its own broadcast domain and routing boundary.

Router Configuration

Each router was configured with:

- Correct IP addresses
- Correct subnet masks
- Interfaces set to **up/up**
- RIP v2 enabled

RIP v2 Networks

Code

192.168.10.0

192.168.20.0

192.168.30.0

Routers learned all networks dynamically, enabling multi-hop routing.

Switch Configuration

Each LAN uses a dedicated switch.

Critical VLAN Fix

During troubleshooting, one switch port (Switch2 → PC2) was found in **VLAN 10**, isolating PC2 from Router2.

Correct configuration:

Code

VLAN: 1

Port Status: On

Link: Up

All switch ports must remain in VLAN 1 for this lab.

This was the main cause of PC2 connectivity failure.

Connectivity Testing

Expected Successful Pings

From **PC0**:

Code
ping 192.168.20.10
ping 192.168.30.10

From **PC1**:

Code
ping 192.168.10.10
ping 192.168.30.10

From **PC2**:

Code
ping 192.168.10.10
ping 192.168.20.10

Troubleshooting Method

To isolate failures:

1. Ping from PC → Router gateway
2. Ping from Router → PC
3. Ping across routers
4. Check switch port VLAN
5. Check interface status
6. Check routing table entries

This method identified the VLAN mismatch as the root cause.

Issues Encountered

1. PC2 unreachable

- Router2 could not ping PC2
- PC0 and PC1 could not reach PC2
- RIP routing was correct
- IP addressing was correct
- Interfaces were up

Root cause: Switch2 port for PC2 was set to **VLAN 10** instead of **VLAN 1**.

2. Multi-hop routing confusion

Understanding how packets travel:

Code

PC0 → Router0 → Router1 → Router2 → PC2

This helped confirm routing was correct and the issue was at Layer 2.

Learning Outcomes

Day 65 builds real engineering skills:

- Designing multi-router networks
- Understanding broadcast domains
- Using RIP for dynamic routing
- Diagnosing VLAN isolation
- Multi-hop troubleshooting
- Verifying routing tables
- Understanding how routers and switches interact

This is the first lab that feels like a real enterprise network.

How This Applies to Real Jobs

This lab directly maps to:

- **Cloud engineering (AWS VPCs, subnets, route tables)**
- **Systems engineering (servers, gateways, VLANs)**
- **Network engineering (Cisco routing/switching)**
- **Cybersecurity (segmentation, lateral movement)**

AI can calculate IPs — but AI cannot troubleshoot VLAN isolation or multi-hop routing. This is the skill companies pay for.

Repository Structure

Code

Day65/

```
|
|— topology.png
|— README.md ← this file
|— notes/
|   |— router-configs.txt
|   |— switch-configs.txt
|   |— troubleshooting-log.txt
```

Next Steps

Choose your direction:

- **Create Day 66 documentation**
- **Convert this into a cloud version using AWS VPCs**
- **Build a portfolio write-up for recruiters**

I can help you polish this even more if you want it GitHub-perfect.